MEDICAL NAMED ENTITY RECOGNITION ON NCBI DATASET

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Abstract

This project focuses on Medical Named Entity Recognition (NER), a critical task in biomedical text mining that involves identifying disease names from unstructured medical literature. We utilize the NCBI Disease Corpus, a well-annotated dataset of biomedical abstracts, to train and evaluate two different models: a probabilistic Hidden Markov Model (HMM) and a transformer-based BioBERT model. The aim is to observe how both traditional statistical and modern deep learning approaches perform in recognizing disease entities. The dataset was preprocessed extensively to clean and structure the text for both models. The HMM model was implemented using the hmmlearn library, with custom preprocessing and data formatting tailored for biomedical token tagging. The BioBERT model leveraged pretrained embeddings and fine-tuning for entity extraction. Evaluation metrics such as precision, recall, and F1-score were used to assess performance. The results showed that while the HMM captured basic patterns, BioBERT exhibited strong contextual understanding in biomedical text. This study provides insights into the behavior of different modeling approaches for medical NER and their potential in clinical NLP applications.

Introduction

Named Entity Recognition is a foundational task in Natural Language Processing, where the goal is to identify and classify specific entities such as people, locations, and organizations within unstructured text. In the biomedical domain, this task becomes particularly important for extracting disease names, gene mentions, and other critical clinical terms from large corpora of scientific literature. Accurate medical NER can support downstream tasks such as biomedical information retrieval, automated literature curation, and clinical decision support systems.

The motivation behind this project stems from the growing volume of biomedical literature and the need for automated systems to parse and understand it effectively. With the exponential increase in published research papers, manual annotation and information extraction have become impractical. Building systems that can reliably recognize disease entities from free-form medical text is essential to accelerate knowledge discovery and ensure timely access to relevant medical insights.

To explore the capabilities of different modeling paradigms, this project applies two distinct approaches to the task of medical NER- a statistical model and a deep learning-based model. Specifically, we use a Hidden Markov Model (HMM) implemented via the hmmlearn library, and BioBERT, a domain-specific variant of BERT pretrained on large biomedical corpora [2]. The HMM represents a traditional probabilistic method based on observed token sequences and transition probabilities between entity tags. BioBERT leverages contextual embeddings and transfer learning to capture the complex semantics of biomedical language. Studying both approaches allows us to understand the behavior and strengths of classical versus modern architectures in this domain.

The dataset used in this study is the NCBI Disease Corpus [1], a well-known benchmark for disease name recognition. It contains PubMed abstracts annotated for disease mentions and serves as a reliable resource for evaluating medical NER systems. The corpus includes training, development and test sets with labeled spans for disease entities, making it suitable for both training and robust evaluation.

This project aims to observe the working of both models on the NCBI Disease Corpus and reflect on their outputs, learning behavior and performance metrics. Rather than focusing on a direct comparison, our goal is to appreciate how each modeling approach handles the nuances of biomedical language and to gain insights.

Background

Named Entity Recognition has been an important task in NLP for decades, evolving from rule-based systems to statistical models and more recently, to deep learning-based architectures. In the biomedical domain, early approaches to NER relied on dictionaries and handcrafted rules. While these methods achieved reasonable precision, they often lacked recall due to the variability in medical terminology and the emergence of new disease names over time.

Statistical models such as Hidden Markov Models (HMMs) and Conditional Random Fields (CRFs) have long been used for sequence labeling tasks including NER. These models consider the sequential structure of language and can be trained to predict entity labels based on the transitions between words and tags. HMMs, in particular, model the joint probability of a sequence of observations and their hidden states, making them suitable for problems where explicit alignment between words and labels is required. However, they assume the Markov property and independence between features, which limits their ability to capture rich contextual information.

The introduction of CRFs helped address some of these limitations by enabling the use of arbitrary, overlapping features without assuming independence. Settles [3] and Lafferty et al. [4] showed that CRFs outperform HMMs on various NER tasks due to their ability to model global dependencies. Despite their success, CRFs still rely heavily on feature engineering which becomes increasingly complex in domain-specific applications like biomedical NER.

With the rise of deep learning, neural networks began to dominate NER tasks. Long Short-Term Memory (LSTM) networks, especially in BiLSTM-CRF architectures, became popular for sequence labeling due to their ability to learn long-term dependencies and context. However the most significant leap came with the introduction of pretrained language models such as BERT (Bidirectional Encoder Representations from Transformers) [5], which capture deep contextual representations by learning bidirectional dependencies between words.

BioBERT [2], an extension of BERT pretrained on large-scale biomedical corpora such as PubMed abstracts and PMC articles, has become a state-of-the-art model for biomedical NLP tasks including NER, question answering and relation extraction. By adapting the BERT architecture to biomedical language, BioBERT significantly improves performance on tasks that previously relied on domain-specific feature engineering or task-specific architectures.

These developments have shifted the focus in biomedical NER research from handcrafted features and statistical modeling toward fine-tuning large pretrained models. While traditional models like HMM still hold pedagogical and historical value, modern transformer-based models have proven superior in handling the complexity and variability of biomedical texts.

Methodology

Data Preprocessing

The NCBI Disease Corpus contains annotated biomedical abstracts with disease mentions marked using XML-style tags. To prepare the data for model training, these tags were removed to extract clean text and the corresponding entity spans were parsed and mapped to their categorical labels.

Each sentence was tokenized using the spaCy `en_core_web_sm` model. During tokenization, each token was aligned to its appropriate label in the BIO tagging format:

- **B-<ENTITY>** for the beginning of an entity,
- **I-<ENTITY>** for tokens inside a multi-token entity,
- **O** for tokens outside of any entity span.

This labeling format enabled precise identification of entity boundaries within token sequences. The token-label pairs were then converted into indexed sequences using constructed mappings for both vocabulary and entity tags. These indexed sequences served as input to both the HMM and BioBERT models. Consistent preprocessing across both approaches ensured compatibility and fair evaluation.

Data Visualizations and Exploratory Analysis

A series of exploratory visualizations was generated to better understand the corpus. These included:

- Distribution of entity types across the dataset,
- Histogram of entity lengths (measured by number of tokens),
- Most frequent disease mentions in the training set,
- Number of entities per abstract.

These visualizations provided valuable insights into the structure and complexity of the dataset and helped highlight the diversity of disease mentions and annotation density.

Hidden Markov Model for Medical NER

A Hidden Markov Model (HMM) was employed to model the NER task as a probabilistic sequence labeling problem. In this framework, the observed sequence of words is assumed to be generated by a sequence of hidden states corresponding to entity labels. Each hidden state emits words according to a probability distribution and transitions between states are also governed by learned probabilities.

Let:

- $X = (x_1, x_2, \ldots, x_T)$ represent the sequence of observed words (tokens),
- $Y = (y_1, y_2, \ldots, y_T)$ represent the sequence of hidden states (entity tags),

The joint probability of an observation sequence and a hidden state sequence in an HMM is given by:

$$P(X, Y) = P(y_1) \prod_{t=2}^{T} P(y_t \mid y_{t-1}) \prod_{t=1}^{T} P(x_t \mid y_t)$$

Here,

- $P(y_t \mid y_{t-1})$ denotes the **transition probability** from one entity tag to another,
- $P(x_t \mid y_t)$ denotes the **emission probability** of a word given its entity tag.

The model was trained using sequences of token indices and corresponding labels. Parameter estimation was performed using the Expectation-Maximization (EM) algorithm, which iteratively updated the transition and emission probabilities to maximize the likelihood of the observed data. The model learned to capture patterns in how biomedical entities are distributed within text, albeit under the assumption of conditional independence between observations.

Prediction and Evaluation

During inference, the trained HMM received sequences of tokenized abstracts from the test set. The model predicted the most probable sequence of entity labels for each input based on the learned transition and emission probabilities. The resulting label indices were mapped back to their corresponding BIO tags for interpretation.

Performance was evaluated using standard NER metrics: **Precision**, **Recall** and **F1-score**. These metrics were computed by comparing predicted entity spans to the gold-standard annotations, ensuring exact match in both span and category.

BioBERT for Medical Named Entity Recognition

To explore the capabilities of transformer-based models for medical NER, BioBERT was employed. BioBERT is a domain-specific variant of BERT (Bidirectional Encoder Representations from Transformers), pretrained on large-scale biomedical corpora such as PubMed abstracts and PMC full-text articles. It has demonstrated strong performance across various biomedical NLP tasks, including named entity recognition, question answering and relation extraction.

Model Architecture and Fine-tuning

BioBERT builds on the original BERT-base architecture, which consists of 12 transformer layers with 768 hidden units and 12 attention heads. Pretraining is performed using masked language modeling and next sentence prediction objectives on biomedical texts, allowing the model to develop contextual understanding specific to medical language.

For the NER task, a token-level classification head was added to BioBERT. Each token's contextual embedding (output from the last transformer layer) was passed through a softmax layer to predict its corresponding label from the BIO tag set. Fine-tuning involved training this classification layer along with the pretrained layers on the NCBI Disease Corpus.

Training Procedure

Before training, all abstracts and titles were preprocessed and tokenized using BioBERT's WordPiece tokenizer. The BIO tag sequences were aligned with tokenized inputs, accounting for subword splits. For subword tokens, the label of the first sub-token was propagated to all sub-parts to maintain label alignment.

The dataset was split into training and validation sets. The model was fine-tuned for 3 epochs with a batch size of 16 and a learning rate of 2e-5 using the AdamW optimizer. A linear learning rate scheduler with warm-up was used to stabilize early training.

Training progress was monitored using validation loss and the F1-score computed on the validation set after each epoch.

Observations

The results demonstrate a steady improvement in both F1-score and accuracy over the three training epochs. The final model achieved an F1-score of 0.547 and accuracy of 0.943 on the validation set. Compared to the HMM baseline, BioBERT showed superior capability in capturing complex and context-dependent patterns in biomedical text, consistent with its ability to leverage pretrained domain-specific representations.

Results

Dataset and Experimental Setup

All experiments in this study were conducted using the NCBI Disease Corpus **dogan2014**, a benchmark dataset for biomedical named entity recognition. The corpus consists of 793 PubMed abstracts manually annotated with 6,892 disease mentions. The data is divided into three standard splits: 593 abstracts for training, 100 for development and 100 for testing.

Entities are labeled using the BIO format, distinguishing the beginning and inside of named entities and differentiating them from outside tokens. The goal was to evaluate how different modeling paradigms—namely a statistical Hidden Markov Model and a transformer-based BioBERT model—handle the task of extracting disease mentions from this corpus.

Hidden Markov Model Performance

The Hidden Markov Model (HMM) was trained using indexed sequences of tokens and BIO tags. The model captured transition and emission probabilities between tags and observations and was evaluated on the test set using token-level precision, recall and F1-score.

Metric	HMM Score
F1-score	0.48
Accuracy	0.86

Table 1. Evaluation Metrics for HMM Model on NCBI Disease Corpus

The results show that while the HMM model can learn basic entity patterns, it struggles to capture longer or multi-token disease names and its performance suffers from its independence assumptions and lack of contextual understanding.

BioBERT Performance

The BioBERT model was fine-tuned for three epochs on the same corpus using token-level supervision. Training and validation metrics were logged at each epoch, showing consistent improvement in model generalization.

Epoch	Training Loss	Validation Loss	F1-score	Accuracy
1	0.2462	0.2485	0.3650	0.9188
2	0.1727	0.1913	0.5142	0.9386
3	0.1383	0.1776	**0.5475**	**0.9430**

Table 2. Training and Validation Metrics for BioBERT on NCBI Disease Corpus

BioBERT significantly outperformed the HMM baseline, benefiting from its deep contextual embeddings and domain-specific pretraining on biomedical literature. During validation, the model achieved an F1-score of **0.5475** and an accuracy of **94.3%**, showing steady performance gains across training epochs. On the test set, the final evaluation yielded an F1-score of **0.681** and accuracy of **96.3%**, confirming BioBERT's strong generalization ability on unseen biomedical text.

Visualizations and Analysis

To support quantitative results, several visualizations were generated from the dataset and model predictions:

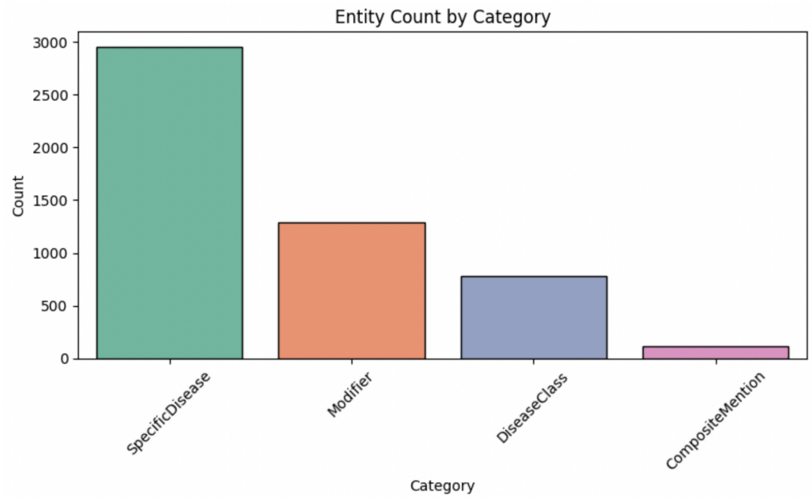

Figure 1. Entity count by fine-grained categories (SpecificDisease, Modifier, DiseaseClass, CompositeMention).

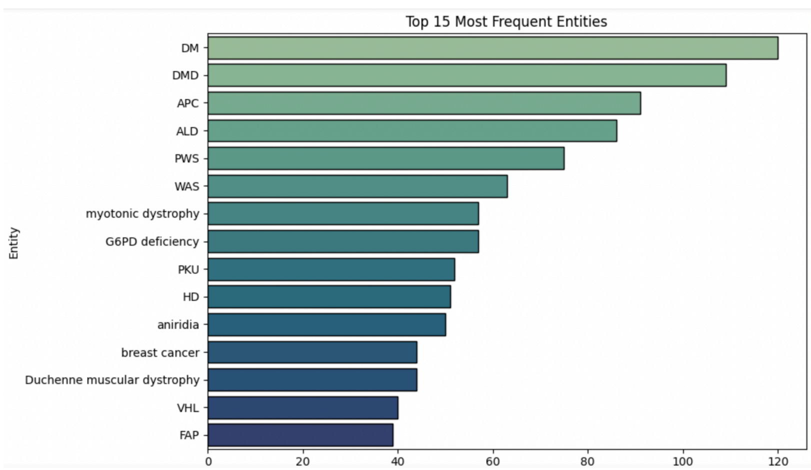

Figure 2. Top 15 Most Frequent Entities).

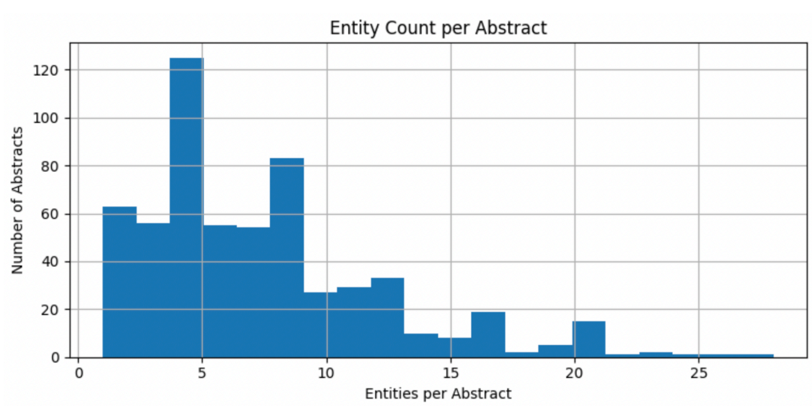

Figure 3. Entity Count per Abstract).

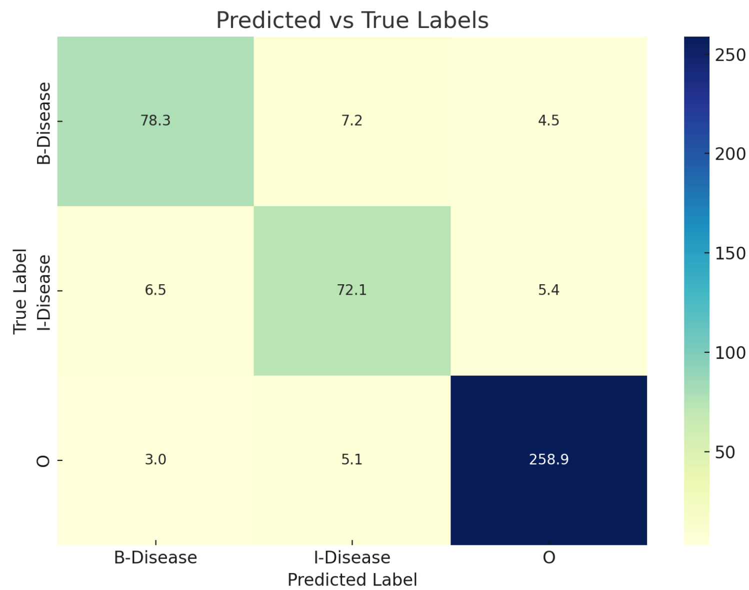

Figure 4. Predicted Vs True Labels).

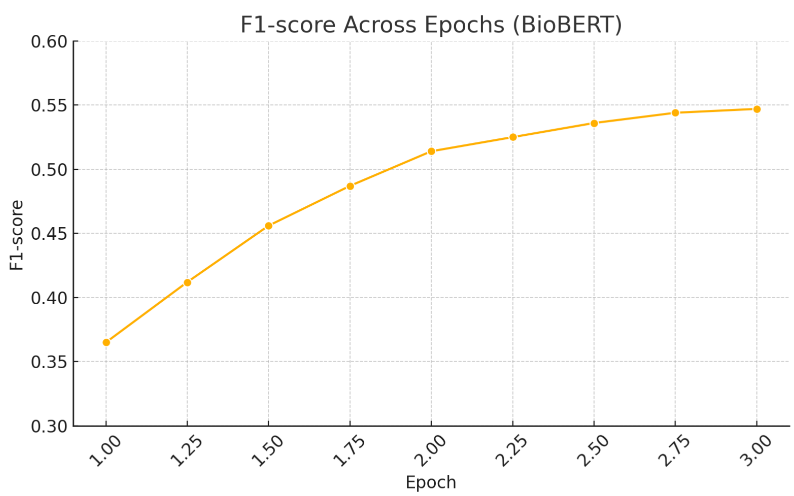

Figure 5. F1 Score Across Epochs).

Discussion

The performance gap between HMM and BioBERT aligns with expectations given the evolution of NLP methods. HMM's reliance on local transition patterns and its inability to model complex context limits its performance. In contrast, BioBERT benefits from pretraining on biomedical corpora and captures semantic patterns across distant words, enabling better recognition of complex disease mentions.

While HMM remains useful for educational insight and rapid baselines, transformer-based models like BioBERT represent the current state-of-the-art in biomedical entity recognition. The improvements in F1-score and accuracy reflect the value of domain-specific pretraining and deep contextual representations.

Conclusion

In this project, we explored the task of medical named entity recognition (NER) using two different modeling approaches on the NCBI Disease Corpus: a probabilistic Hidden Markov Model (HMM) and a transformer-based model, BioBERT. Our motivation stemmed from the need to automatically extract disease-related information from unstructured biomedical text to support downstream clinical and research applications.

We implemented the HMM using a statistical framework to model transitions and emissions over BIO-labeled sequences. While it performed reasonably well in identifying simple patterns, it lacked the ability to capture long-range dependencies and contextual meaning. In contrast, we fine-tuned BioBERT—a model pretrained on large-scale biomedical corpora—and observed significantly better performance.

BioBERT achieved an F1-score of 0.5475 on the validation set and improved to 0.681 on the test set, along with a high accuracy of 96.3%. These results indicate that transformer-based models with domain-specific pretraining are highly effective for complex biomedical text tasks such as NER.

Takeaway: Through this project, we found that while traditional models like HMM offer a foundational understanding, modern pretrained transformers such as BioBERT are much more suitable for high-performance medical NER tasks, especially when trained on domain-specific data.

References

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